EAST WEST UNIVERSITY

CSE489: Mobile Application Development

Spring 2025

Section: 02

Lost And Found App

Submitted to:

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Submitted by:

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Date: 21.05.2025

Work Responsibility

Our Lost and Found Android application was developed collaboratively by three team members, with each of us responsible for specific modules to ensure a well-structured and functional product.

Tasnuva Islam Mila - Authentication System:

- ➔ Implemented Firebase Authentication for Email/Password and Google Sign-In.
- ➔ Handled user registration, login, and secure session management.
- ➔ Integrated terms and conditions check and error validations for better user experience.
- ➔ Managed Firestore user data creation upon signup.

Md. Asifuzzaman - Profile Management:

- ➔ Designed and implemented the user profile section.
- ➔ Enabled users to view, edit, and update their personal information.
- ➔ Ensured changes reflect in real-time using Firestore.
- ➔ Managed profile UI using both XML and Java.

Nayma Amin Nishy - Post Handling & Management:

- ➔ Built functionality for users to create, view, edit, and delete lost/found item posts.
- ➔ Designed intuitive forms for post submission with fields like category, location, and description.
- ➔ Implemented category and location-based filtering for efficient search.
- ➔ Developed UI and backend logic using XML and Java.

Introduction:

- **Background:** In today's fast-paced world, losing personal belongings has become a common occurrence, especially in public places like schools, colleges, offices, or transport systems. Retrieving lost items often becomes a challenging and time-consuming task due to the lack of a proper system to report or claim such items. To address this issue, we have developed a Lost and Found Android Application using Java in Android Studio.

  The application serves as a centralized platform where users can report lost or found items, view all posted items, and contact relevant users to reclaim or return items. It offers an intuitive interface and leverages Firebase services (Authentication, Firestore, and Storage) for secure user management and real-time data handling.

- **Motivation**: The primary motivation behind building this application is to solve a real-world problem faced by students, staff, and the general public — the inconvenience and frustration of losing and trying to recover personal belongings. Traditional notice boards or verbal communication are inefficient and lack reach.

  We aimed to create a digital solution that:

  - ➢ Allows users to quickly report found items.
  - ➢ Helps users who have lost items to check if someone else has found and reported them.
  - ➢ Ensures smooth and secure communication between the owner and the finder.

By using Firebase for backend operations, we ensure the app is scalable, secure, and maintains real-time synchronization of posts and user activities.

- **Benefits:**
  - ➢ **Efficient Recovery of Items**: Increases the chances of recovering lost items by providing a platform to instantly report and view lost or found items.
  - ➢ **Real-time Updates**: Thanks to Firebase, all updates (new posts, edits, or deletions) are reflected in real-time.
  - ➢ **Secure User Authentication**: Uses Firebase Authentication to ensure that all users are verified before posting or contacting.
  - ➢ **Easy to Use**: A clean and straightforward user interface makes it accessible even for users with minimal technical skills.
  - ➢ **Profile Management**: Users can edit their profiles, update images, and manage their posts from a single dashboard.

- **Usability:**
  - ➢ **Lost or Found Posting**: Users can create posts by selecting the item type (Lost or Found), entering a description, date, location, and uploading an image.
  - ➢ **Tabbed View**: Users can browse posts under separate tabs: All Posts, Lost Items, and Found Items.
  - ➢ **Post Details & Contact**: Clicking on a post reveals its full details along with the contact information if available.

- **User Profile**: Users can view and update their profile, including uploading a profile image and selecting gender.

- **Secure Session Management**: Includes proper login/logout mechanisms with session clearing for data safety.

- **Firebase Integration**: Uses Firebase Firestore for storing post data, Firebase Auth for user management, and Firebase Storage for handling images.

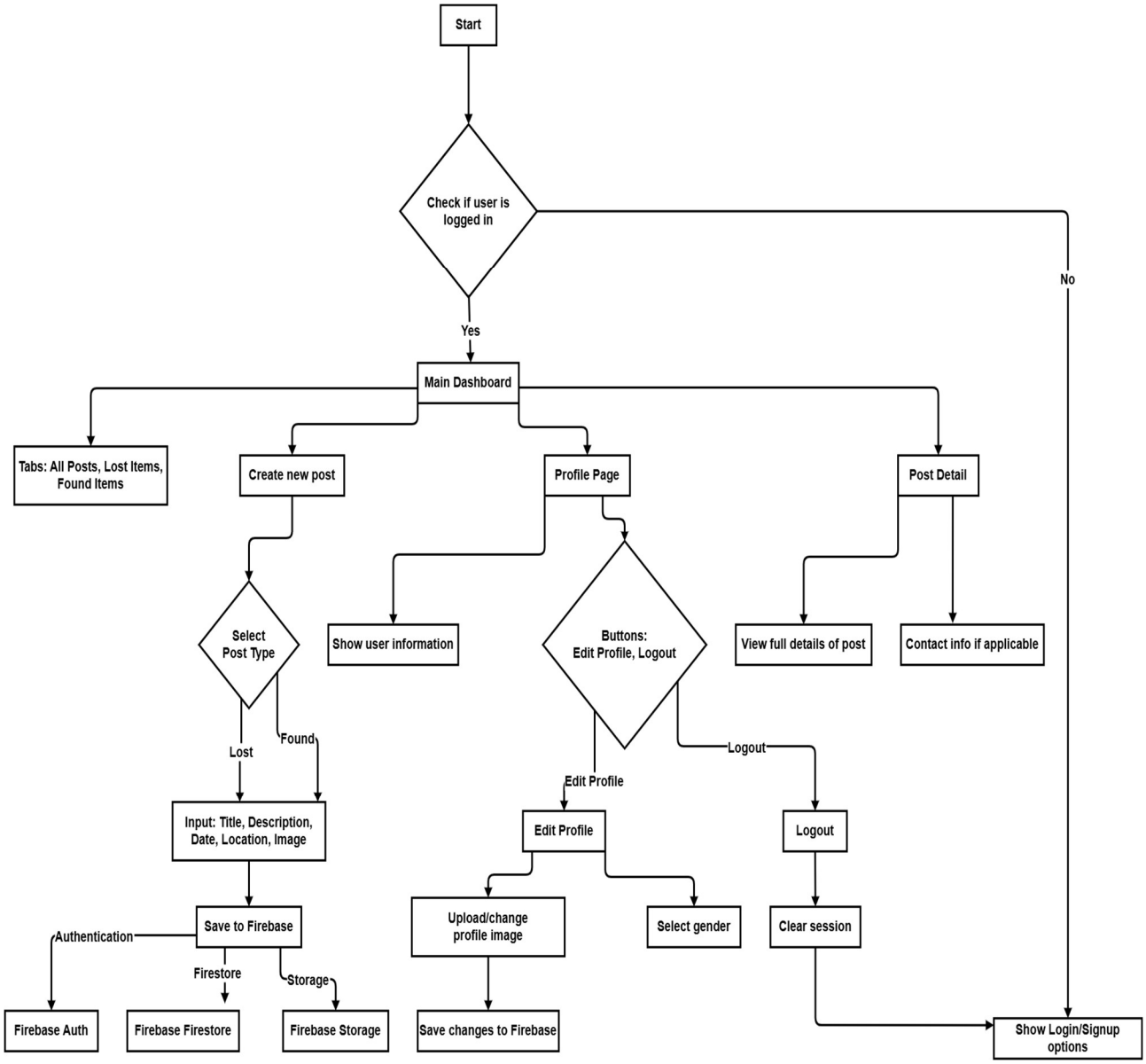
App Overview:

The Lost and Found Android App is a mobile platform developed using Java in Android Studio, aimed at helping users report and recover lost or found items. Integrated with Firebase Authentication, Firestore, and Storage, the app offers secure login, real-time data handling, and cloud-based image uploads.

Users can create posts by selecting the item type (lost/found), entering details, and uploading images. All posts are categorized and displayed under tabs for easy browsing. The app also includes a profile section where users can update their information and manage their posts.

Designed with a simple and intuitive interface, the app promotes quick reporting and recovery of items, making it highly useful in schools, offices, or public settings.

Block Diagram:



Special Feature:

- ➢ **Profile customization with image uploads:** Users can personalize their profiles by uploading a profile picture and editing personal details.
- ➢ **Tabbed filtering for Lost, Found, and All Posts:** Posts are organized under separate tabs for quick access to lost, found, or all items.
- ➢ **Session-based login/logout for enhanced security:** Secure authentication ensures that user sessions are safely managed and protected.
- ➢ **Location input for better item traceability:** Users can enter the location of lost or found items to help others identify nearby posts.
- ➢ **Category-Based Search for Quick Filtering:**Users can filter lost and found posts by specific categories such as Electronics, Documents, Bags, or Pets

Business Model

The Lost and Found Android App is designed primarily as a community service platform to help individuals report and recover misplaced items. However, it also has the potential for growth into a sustainable business through the following models:

1. Freemium Model

The core functionality of the app such as viewing, posting, and contacting users is offered for free. Advanced features can be introduced in the future under a **premium tier**, such as:

- Priority visibility for posts (boosted posts)
- Auto-location tagging using GPS
- Post analytics (views, reach, etc.)
- Ad-free experience

2. In-App Advertising

The app can generate revenue by integrating non-intrusive ads via platforms like **Google AdMob**. Ads can be placed:

- In the main dashboard
- Between post lists
- At the bottom of the profile or post detail screens

This provides a steady income stream without charging users directly, keeping the platform free for everyone.

3. Partnerships with Institutions

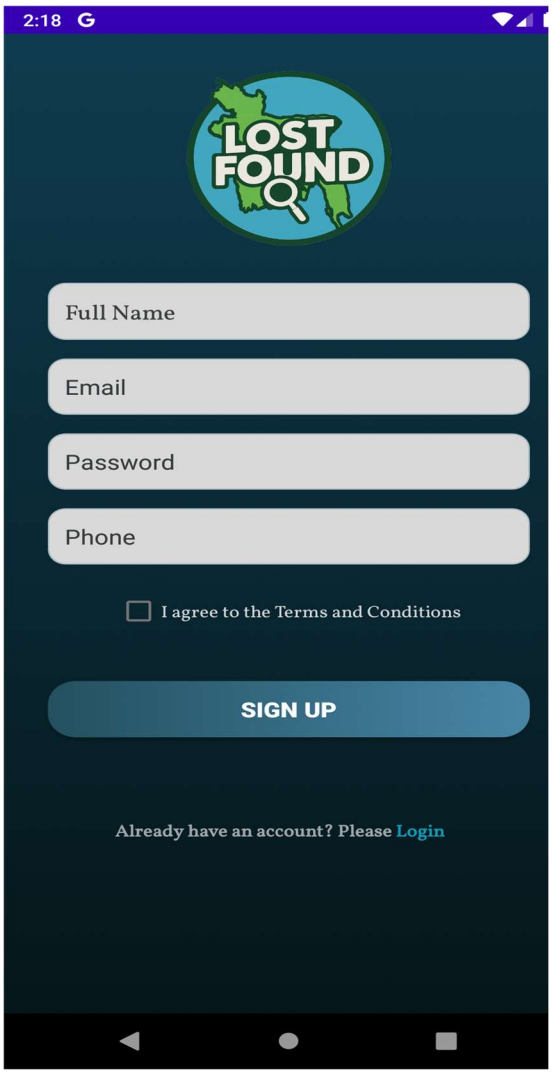
Schools, colleges, offices, or public facilities can **partner with the app** to manage their own internal lost and found systems. A custom dashboard or branded version of the app can be offered as a **subscription-based service**, allowing institutions to track and manage items within their environment.

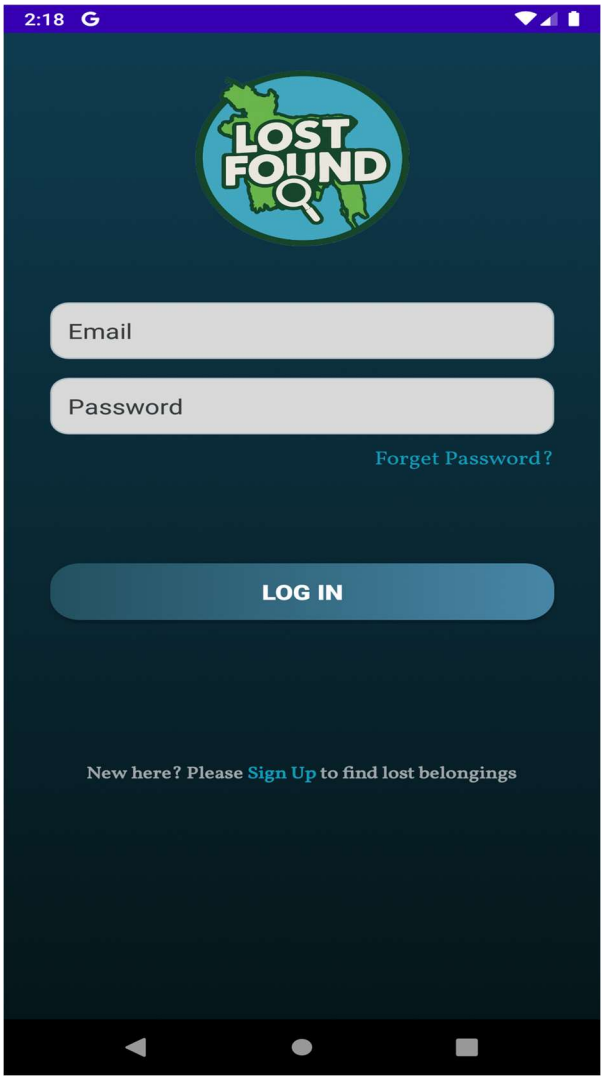
4. Local Business Promotions

The app can offer **location-based promotions** from nearby shops or services (e.g., printing shops, locksmiths, bag sellers) who may benefit from users needing help after losing items. These businesses can pay for sponsored visibility within the app.

Screenshots:

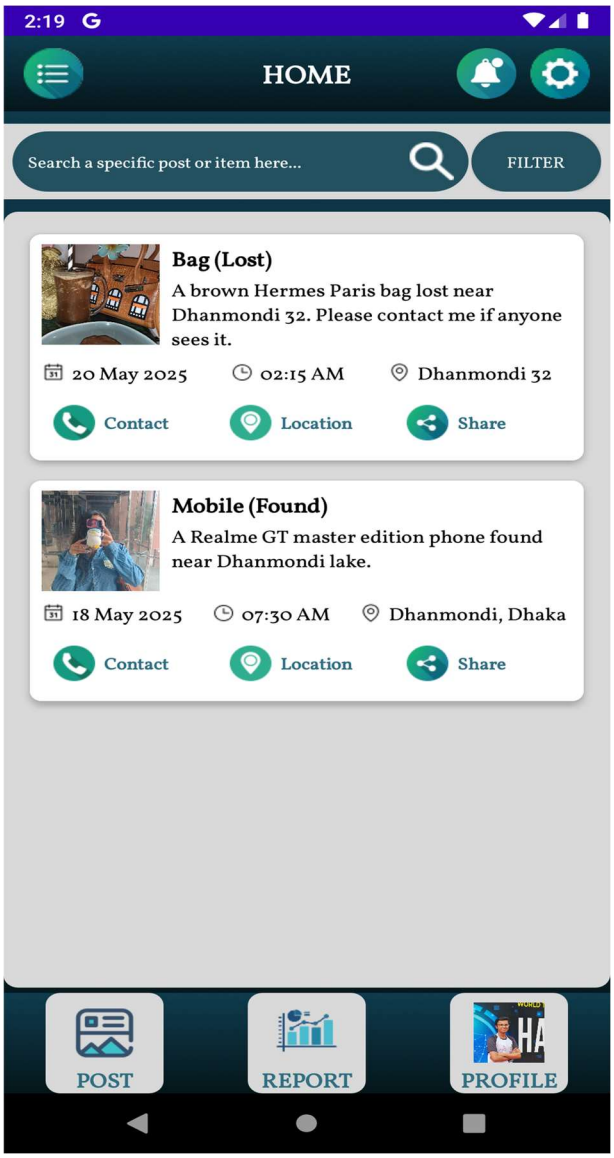
Signup and Login:

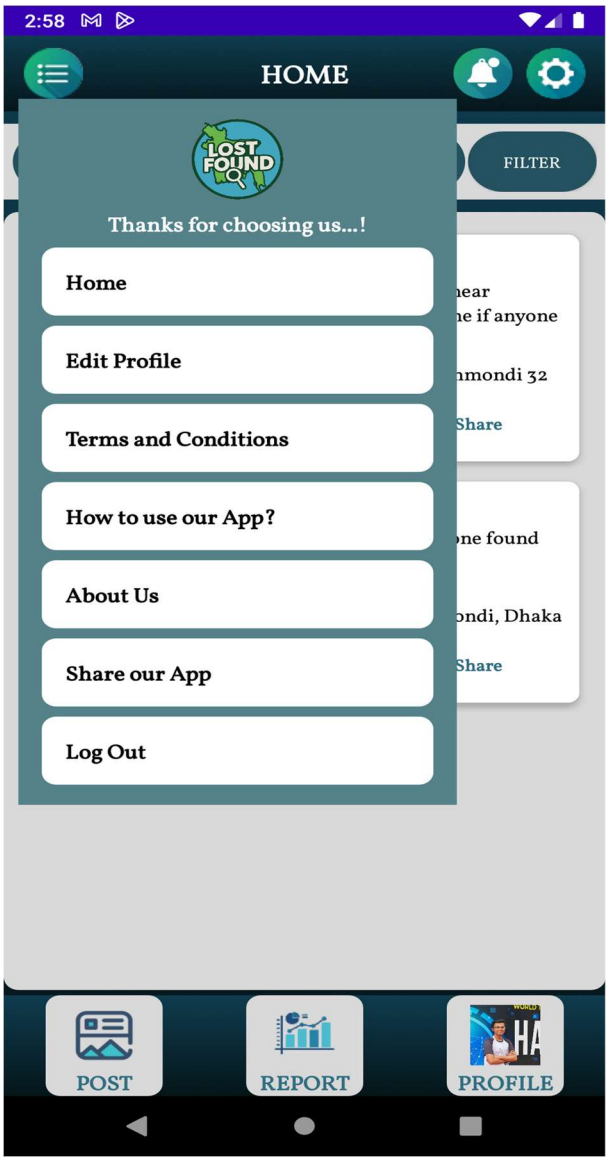




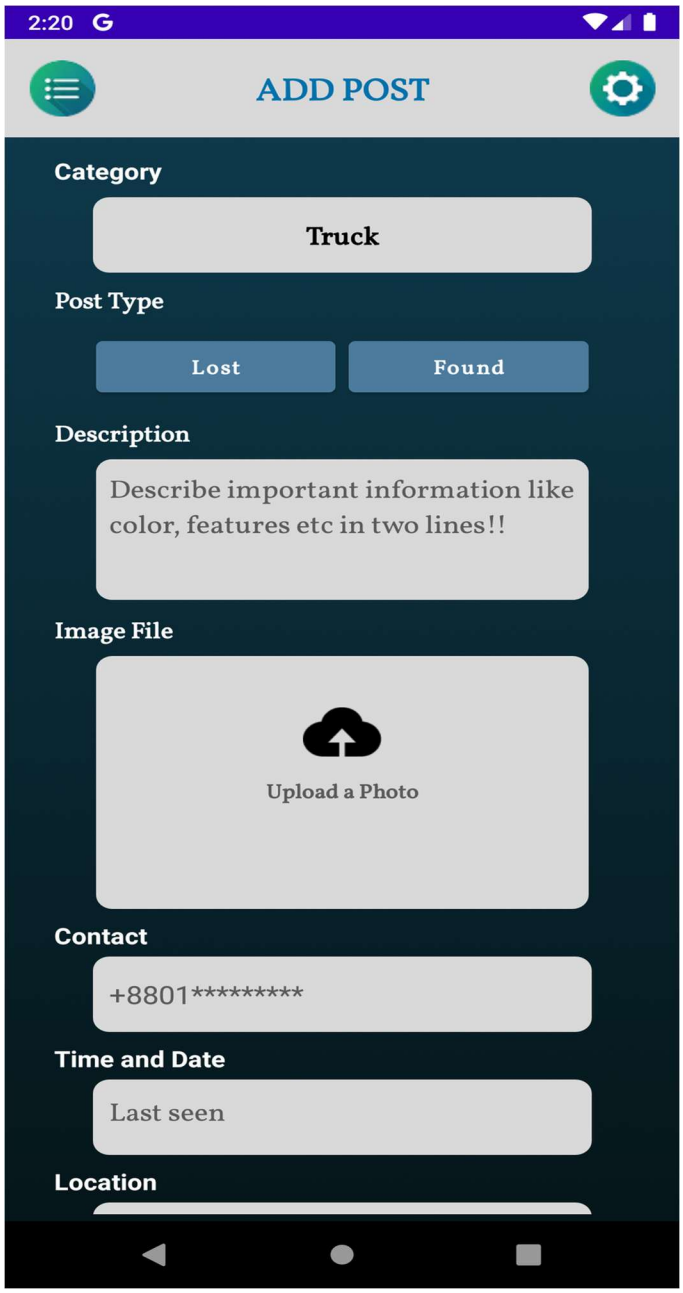
Profile & Edit Profile:

Home & Menu:

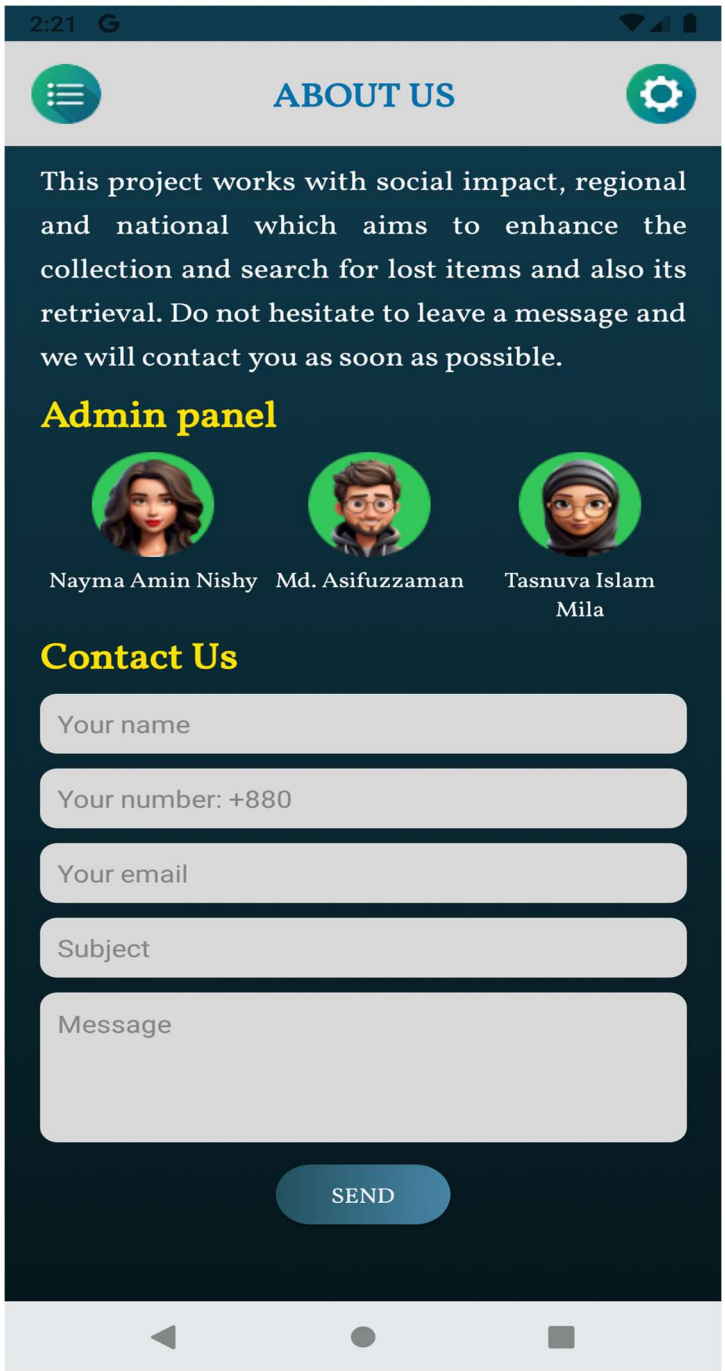

2:19
HOME
Search a specific post or item here...
FILTER
Bag (Lost)
A brown Hermes Paris bag lost near Dhanmondi 32. Please contact me if anyone sees it.
20 May 2025
02:15 AM
Dhanmondi 32
Contact
Location
Share
Mobile (Found)
A Realme GT master edition phone found near Dhanmondi lake.
18 May 2025
07:30 AM
Dhanmondi, Dhaka
Contact
Location
Share
POST
REPORT
PROFILE


2:58
HOME
FILTER
LOST FOUND
Thanks for choosing us...!
Home
Edit Profile
Terms and Conditions
How to use our App?
About Us
Share our App
Log Out
POST
REPORT
PROFILE

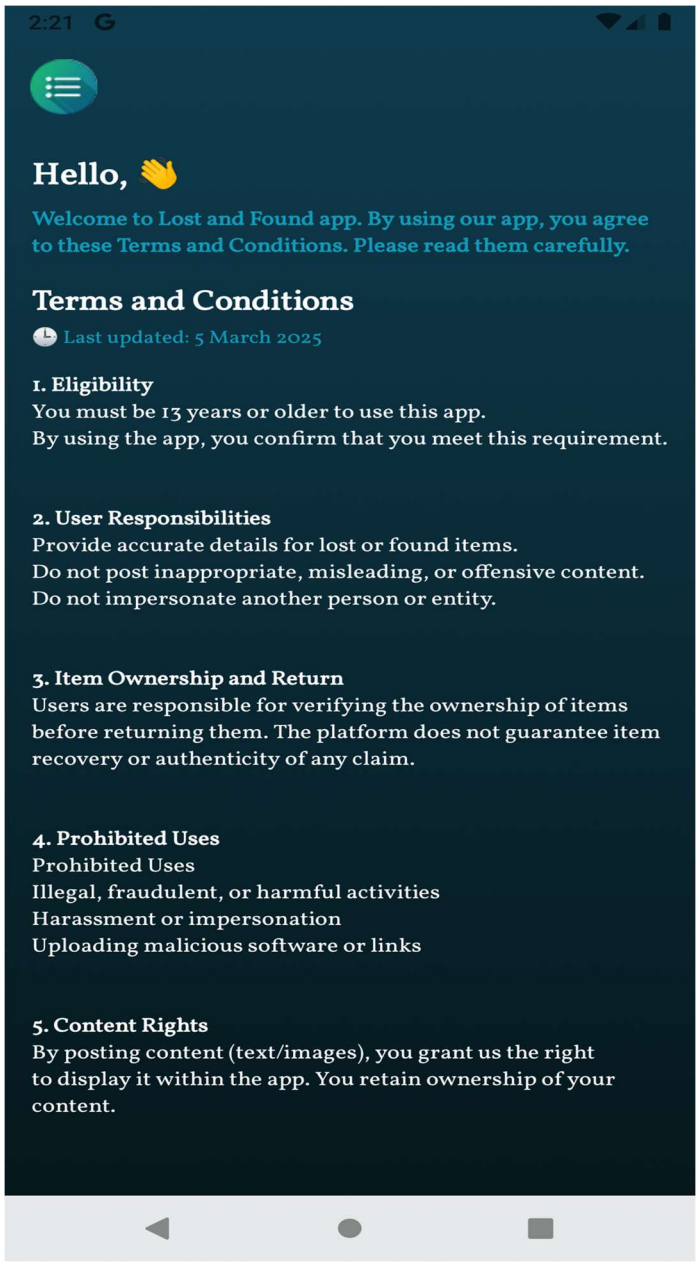
Add Post:


2:20
ADD POST
Category
Truck
Post Type
Lost
Found
Description
Describe important information like color, features etc in two lines!!
Image File
Upload a Photo
Contact
+8801*********
Time and Date
Last seen
Location

About us:


2:21
ABOUT US
This project works with social impact, regional and national which aims to enhance the collection and search for lost items and also its retrieval. Do not hesitate to leave a message and we will contact you as soon as possible.
Admin panel
Nayma Amin Nishy
Md. Asifuzzaman
Tasnuva Islam Mila
Contact Us
Your name
Your number: +880
Your email
Subject
Message
SEND

Terms & Conditions:



2:21

Hello, 👋

Welcome to Lost and Found app. By using our app, you agree to these Terms and Conditions. Please read them carefully.

Terms and Conditions

🕒 Last updated: 5 March 2025

1. Eligibility
You must be 13 years or older to use this app.
By using the app, you confirm that you meet this requirement.

2. User Responsibilities
Provide accurate details for lost or found items.
Do not post inappropriate, misleading, or offensive content.
Do not impersonate another person or entity.

3. Item Ownership and Return
Users are responsible for verifying the ownership of items before returning them. The platform does not guarantee item recovery or authenticity of any claim.

4. Prohibited Uses
Prohibited Uses
Illegal, fraudulent, or harmful activities
Harassment or impersonation
Uploading malicious software or links

5. Content Rights
By posting content (text/images), you grant us the right to display it within the app. You retain ownership of your content.

Challenges Faced During Development

1. Repeated Login During Testing

During development and testing, the Firebase Authentication session often expired or required repeated logins. This slowed down testing and made it time-consuming to verify features that needed user authentication.

2. Dependency Management

Managing various Firebase dependencies (Authentication, Firestore, and Storage) in Android Studio was a challenge. Version mismatches or updates sometimes led to build errors or conflicts with other libraries.

3. Asynchronous Data Handling

Firebase operations such as reading/writing data and image uploads are asynchronous. Handling callbacks properly while keeping the UI responsive required careful planning and debugging, especially in post creation and profile updates.

4. Image Upload and Compression

Uploading images to Firebase Storage required optimization to prevent slow uploads or app crashes due to large file sizes. Implementing image compression before upload was an extra step that added complexity.

5. UI Consistency Across Devices

Ensuring the app UI looked and functioned consistently across different screen sizes and Android versions required a lot of testing and responsive design adjustments.

6. Testing Real-Time Updates

Testing real-time updates for posts (especially in the tabbed view) required frequent syncing and manual refreshing to ensure that the data reflected changes instantly without delays.

Conclusion

The Lost and Found Android App provides an efficient and user-friendly platform for individuals to report and recover lost or found items within a community. With features like profile customization, categorized post filtering, secure login/logout, and location input, the app enhances the chances of successful item recovery. Despite challenges faced during development, such as dependency issues and testing constraints, the project successfully integrates essential services using Firebase to deliver real-time functionality. This app not only addresses a common real-world problem but also showcases the practical application of mobile development skills and cloud integration.